

Operating Systems

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SEC-5

WEEK-5

5. Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Code

```
prudvi@LAPTOP-TEKO376B: ~ × + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int p[2];
    pipe(p);
    if (fork() == 0)
    {
        char s[20];
        close(p[1]);
        read(p[0], s, 20);
        printf("%s\n", s);
    }
    else
    {
        close(p[0]);
        write(p[1], "Hello", 6);
        wait(NULL);
    }

    pipe(p);
    if (fork() == 0)
    {
        dup2(p[1], 1);
        execlp("ls", "ls", "-l", NULL);
    }
    if (fork() == 0)
    {
        dup2(p[0], 0);
        execlp("grep", "grep", ".c", NULL);
    }

    wait(NULL);
    wait(NULL);
    wait(NULL);
    return 0;
}
```

output

```
prudvi@LAPTOP-TEK0376B:~$ nano grep.c
prudvi@LAPTOP-TEK0376B:~$ gcc grep.c
prudvi@LAPTOP-TEK0376B:~$ ./a.out
Hello
-rw-r--r-- 1 prudvi prudvi 503 Sep 1 04:05 Process1.c
-rw-r--r-- 1 prudvi prudvi 632 Sep 15 05:01 child.c
-rw-r--r-- 1 prudvi prudvi 723 Sep 15 04:50 copy.c
-rw-r--r-- 1 prudvi prudvi 0 Aug 4 04:20 floder1.c
-rw-r--r-- 1 prudvi prudvi 256 Aug 4 04:16 floder3.c
-rw-r--r-- 1 prudvi prudvi 319 Aug 11 04:12 floder6.c
-rw-r--r-- 1 prudvi prudvi 182 Aug 11 02:54 folder1.c
-rw-r--r-- 1 prudvi prudvi 255 Aug 11 02:58 folder2.c
-rw-r--r-- 1 prudvi prudvi 277 Aug 11 03:07 folder3.c
-rw-r--r-- 1 prudvi prudvi 165 Aug 11 03:10 folder4.c
-rw-r--r-- 1 prudvi prudvi 218 Aug 11 03:10 folder5.c
-rw-r--r-- 1 prudvi prudvi 319 Aug 11 04:10 folder6.c
-rw-r--r-- 1 prudvi prudvi 614 Sep 15 04:56 fork.c
-rw-r--r-- 1 prudvi prudvi 600 Sep 15 05:09 grep.c
-rw-r--r-- 1 prudvi prudvi 898 Sep 15 04:17 linux.c
-rw-r--r-- 1 prudvi prudvi 502 Sep 1 03:58 peocess1.c
-rw-r--r-- 1 prudvi prudvi 452 Sep 1 03:44 process.c
-rw-r--r-- 1 prudvi prudvi 319 Aug 12 03:55 process.c.save
-rwxr-xr-x 1 prudvi prudvi 16096 Sep 15 03:14 process1
-rw-r--r-- 1 prudvi prudvi 502 Sep 15 03:13 process1.c
-rw-r--r-- 1 prudvi prudvi 505 Sep 1 04:10 process1.c.save
-rw-r--r-- 1 prudvi prudvi 514 Aug 11 04:09 w1.c
```